Linux Essentials

NAVIGATION

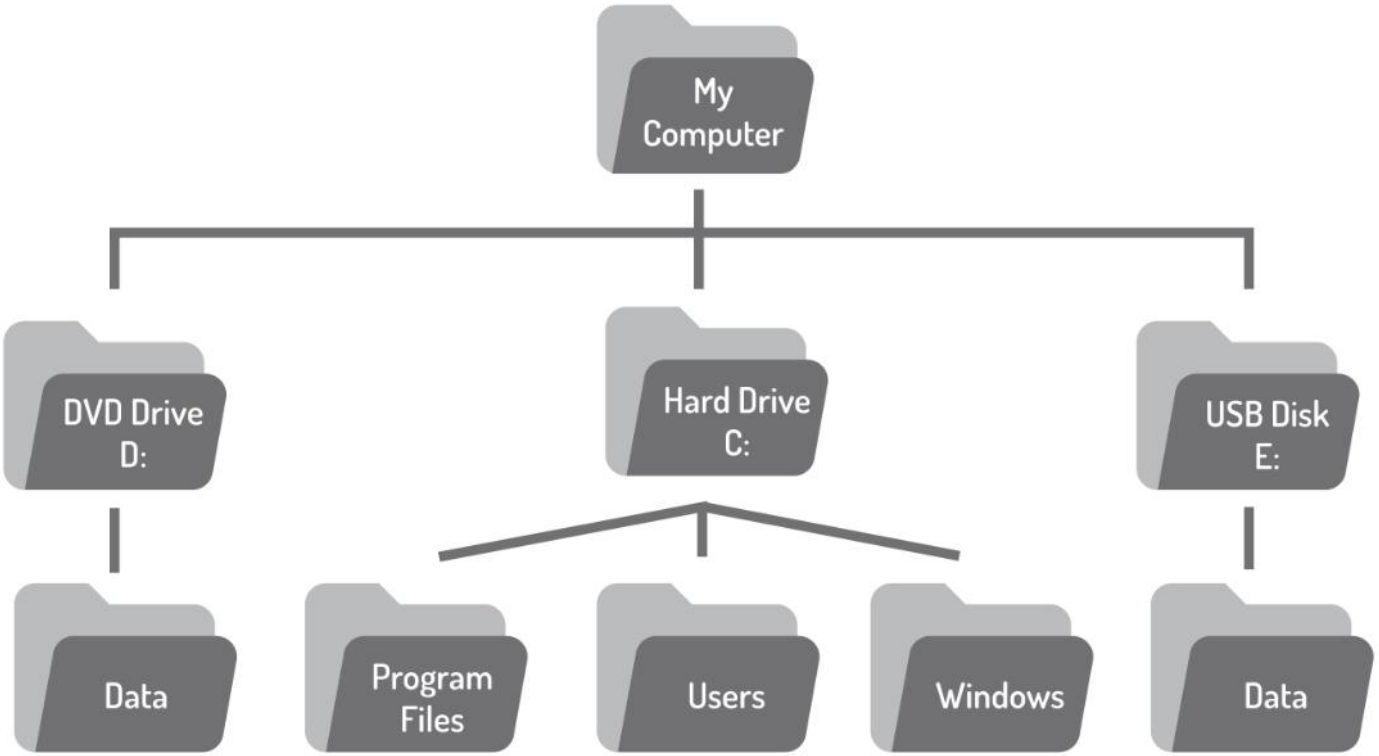
Introduction

- In Linux, everything is considered a **file**.
- ***Files*** are used to store data such as text, graphics, and programs.
- ***Directories*** are a type of file used to store other files; Windows and Mac OS X users typically refer to them as *folders*.
- In any case, directories are used to provide a **hierarchical** organization structure. However, this structure may be somewhat different depending on the type of system in use.

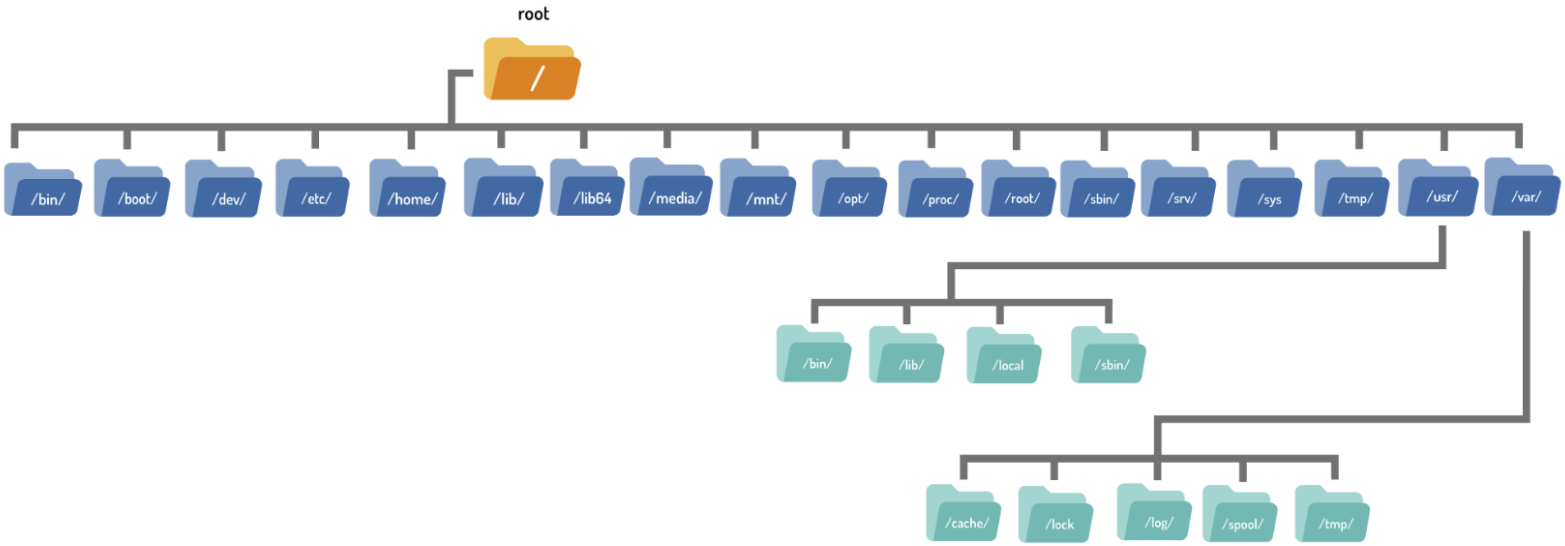
✓ When working in a Linux operating system, it is important to know how to manipulate files and directories.

✓ Some Linux distributions have GUI-based applications that allow you to manage files, but it is advantageous to know how to perform these operations via the command line.

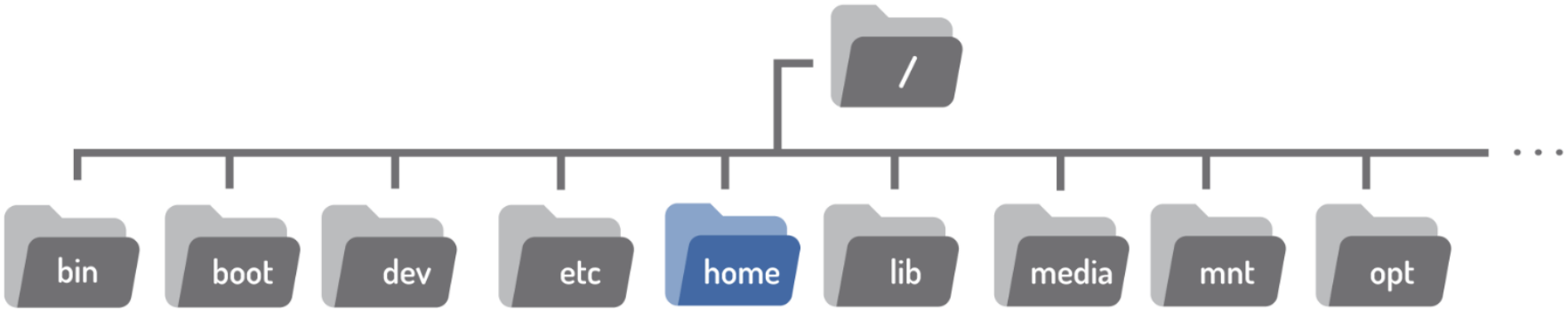
Directory Structure

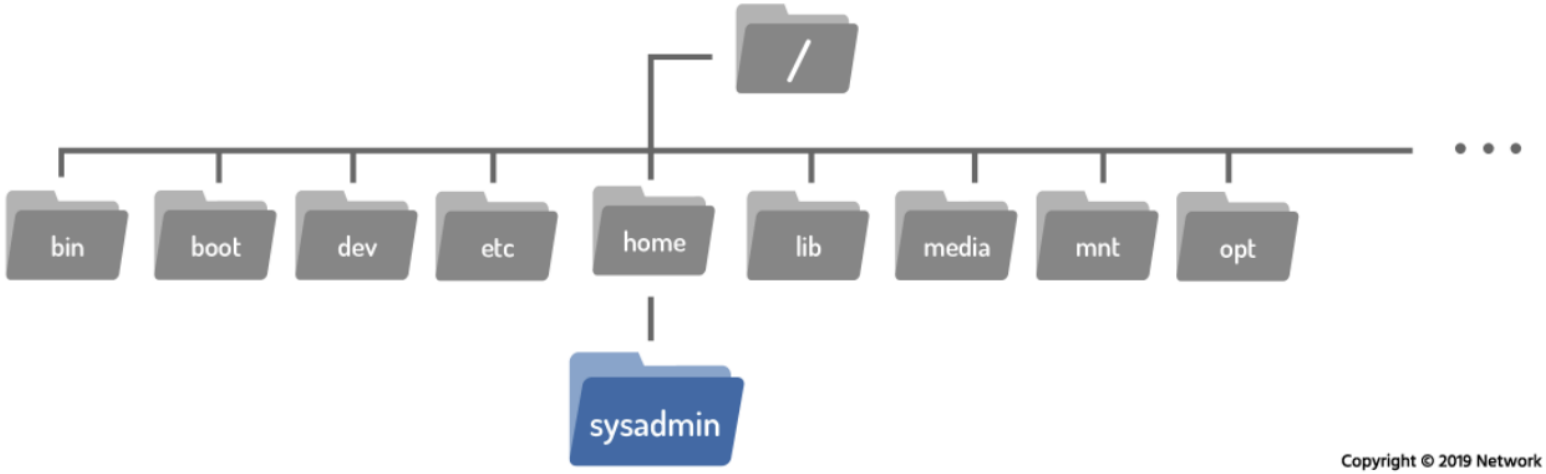


Directory Structure



Home Directory





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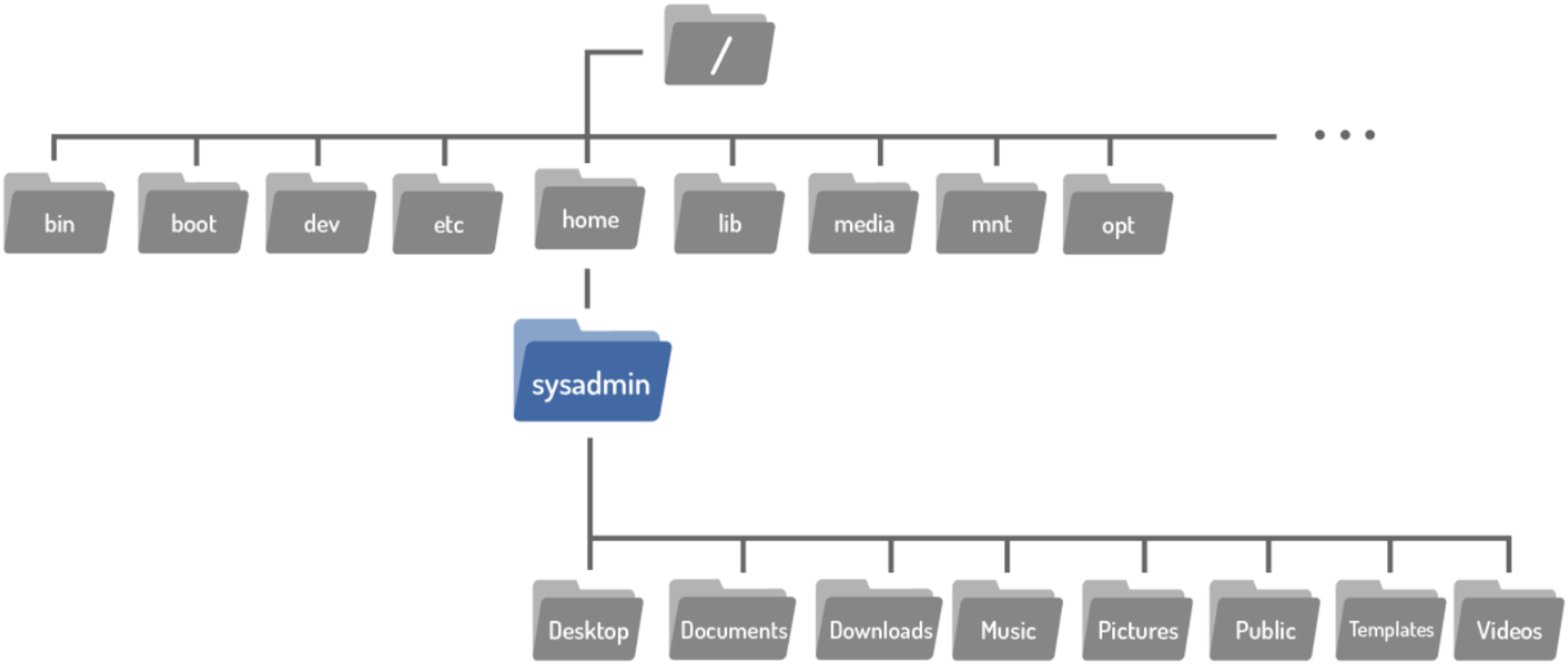
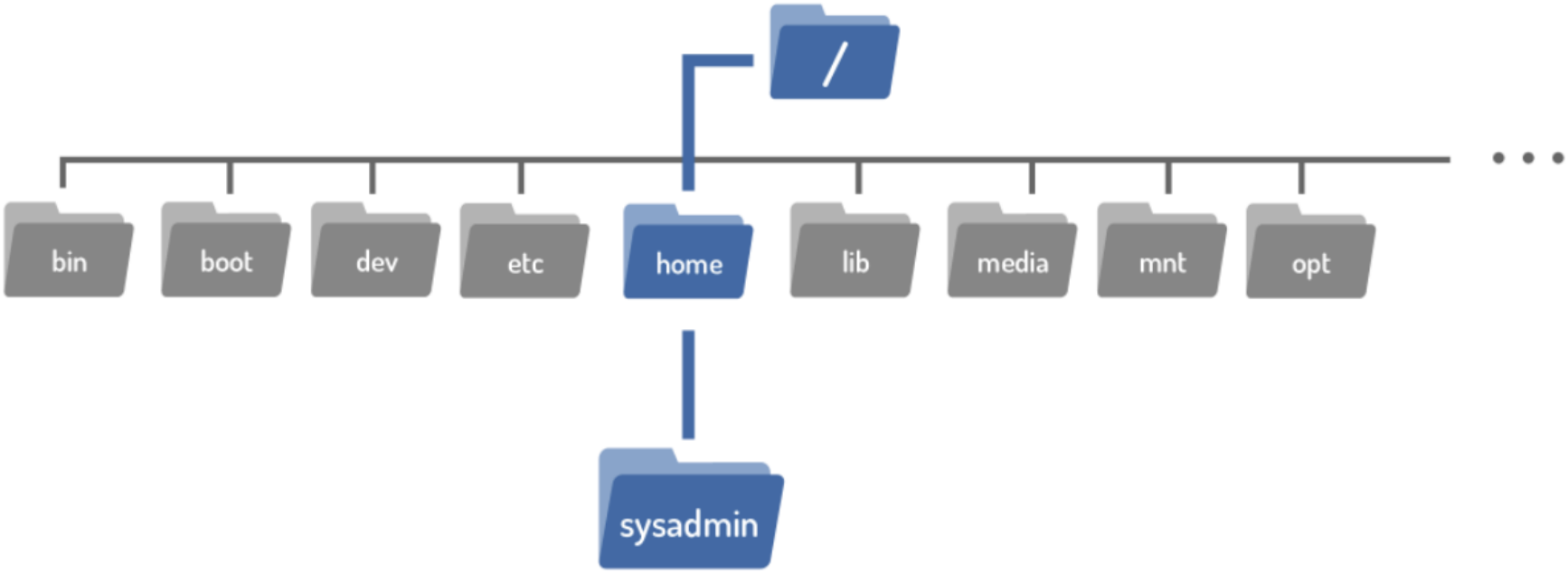
Home Directory

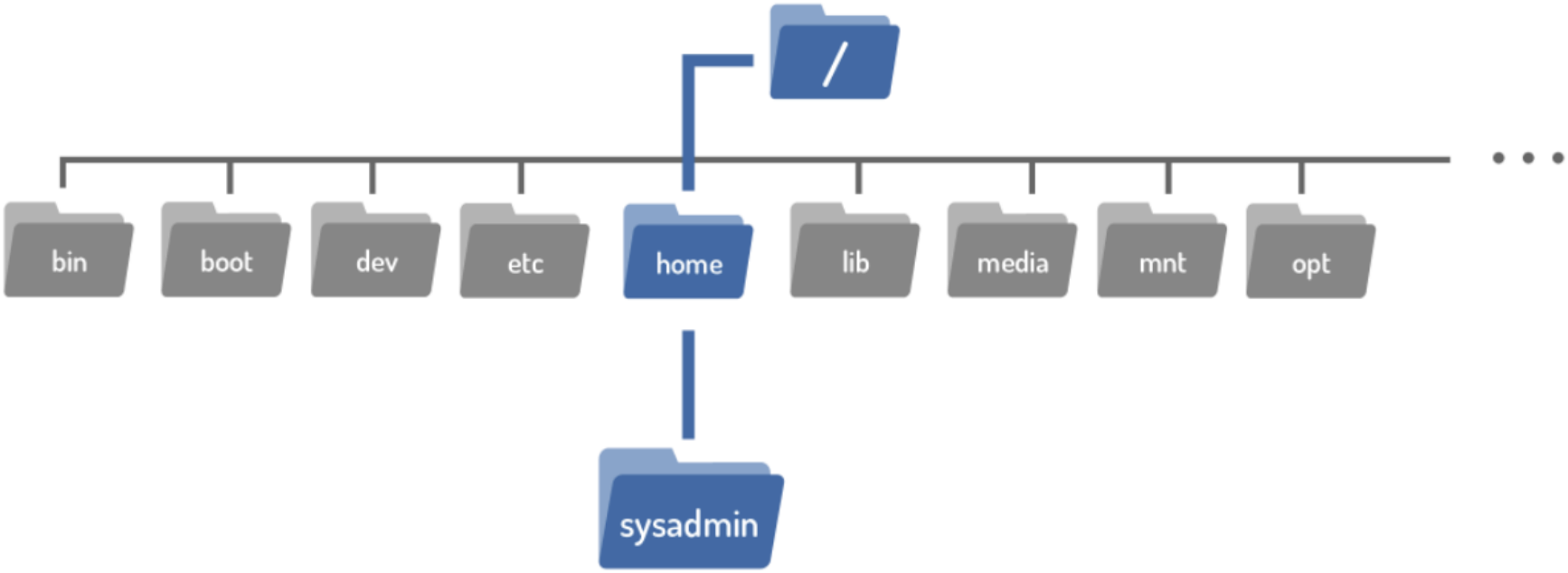


Paths



Path: `/home/sysadmin`

Paths



Path: `/home/sysadmin`

Absolute Paths

```
sysadmin@localhost:~/Documents$ cd /home/sysadmin
```

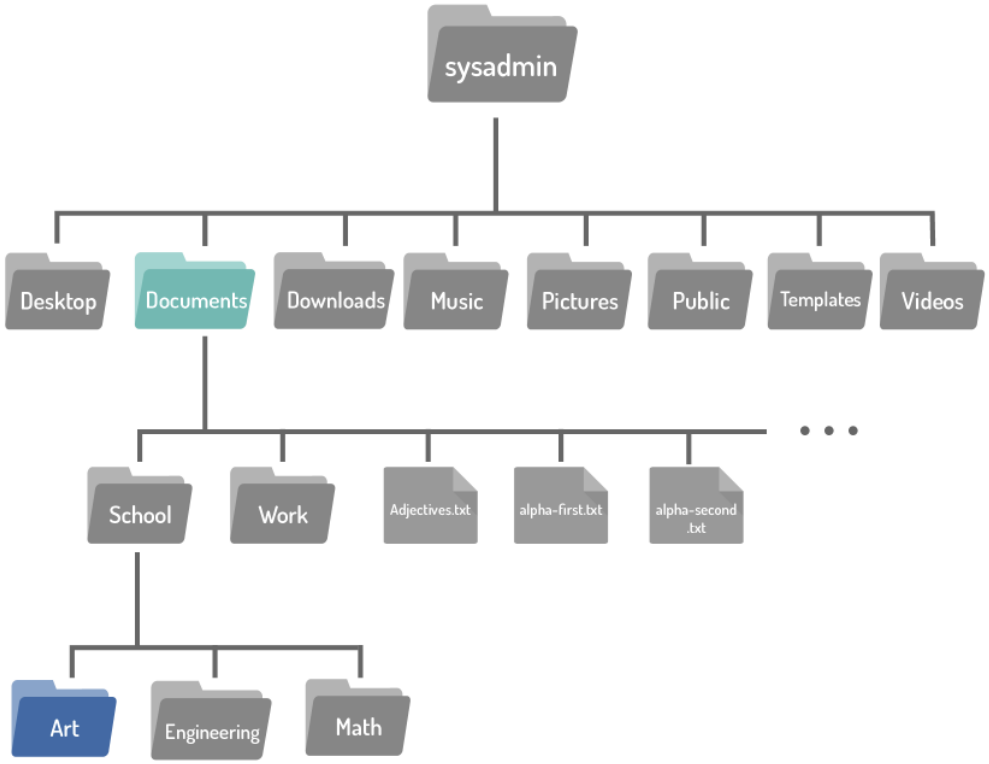
```
sysadmin@localhost:~$ pwd
/home/sysadmin
```

Relative Paths

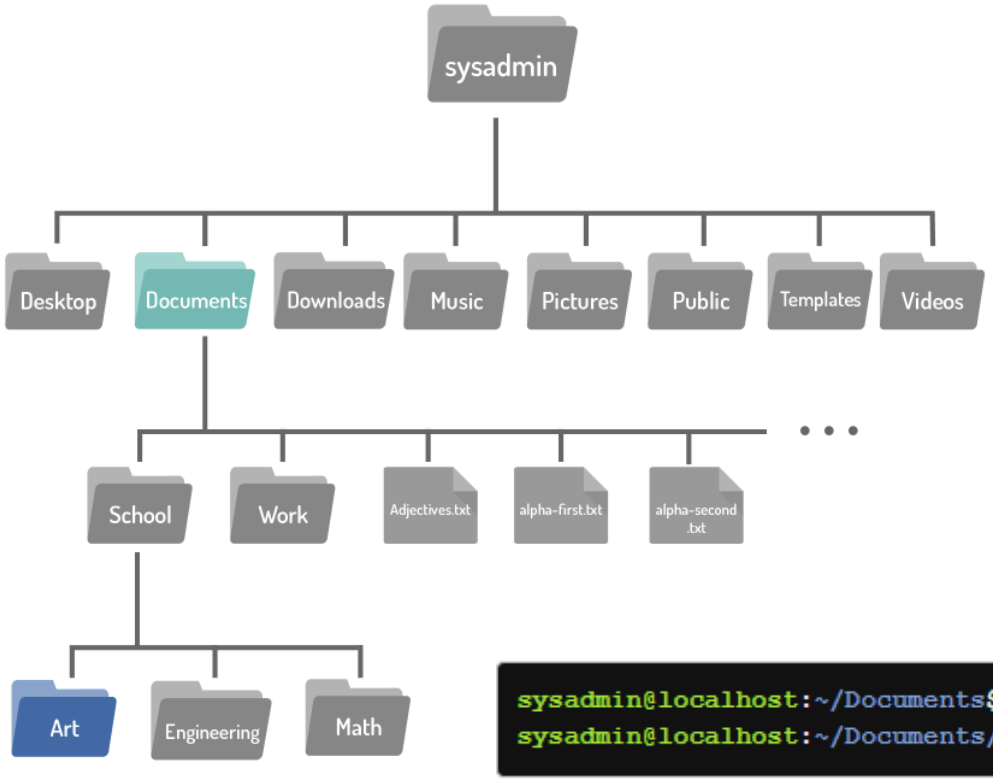
- *Relative paths* start from the current directory.
- A relative path gives directions to a file relative to the current location in the filesystem

```
sysadmin@localhost:~$ cd Documents
sysadmin@localhost:~/Documents$
```

Example:

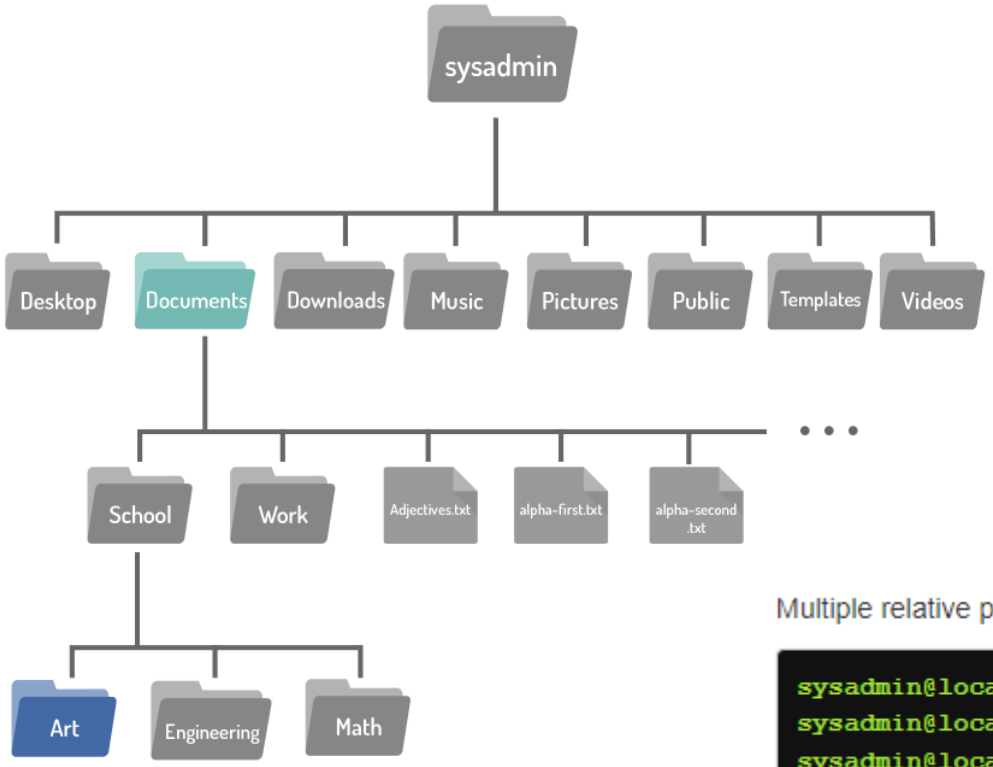

sysadmin
Desktop
Documents
Downloads
Music
Pictures
Public
Templates
Videos
School
Work
Adjectives.txt
alpha-first.txt
alpha-second .txt
Art
Engineering
Math

Example: Absolute Path




```
sysadmin@localhost:~/Documents$ cd /home/sysadmin/Documents/School/Art
sysadmin@localhost:~/Documents/School/Art$
```

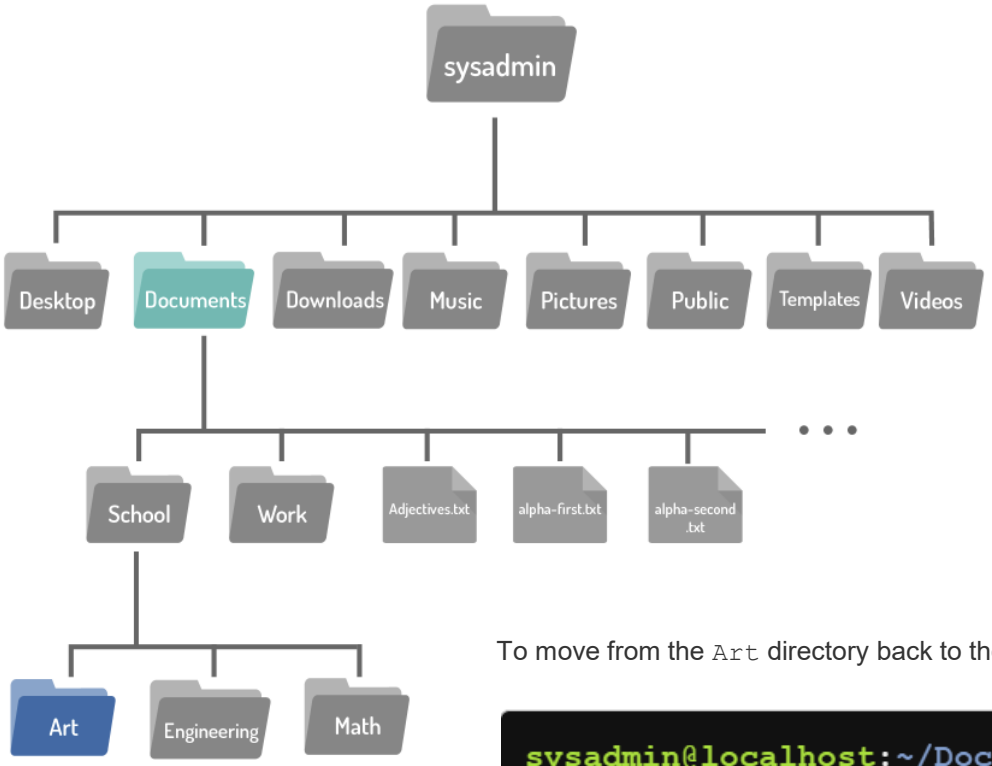
Example: Relative Path



Multiple relative paths can be used:

```
sysadmin@localhost:~/Documents$ cd School
sysadmin@localhost:~/Documents/School$ cd Art
sysadmin@localhost:~/Documents/School/Art$
```

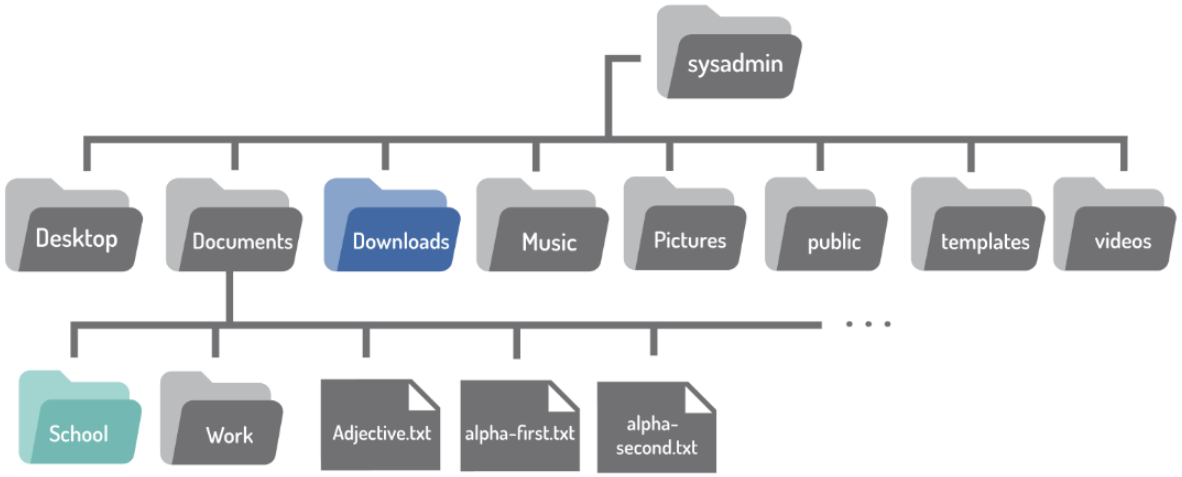
Shortcut: The .. Characters



To move from the `Art` directory back to the `School` directory:

```
sysadmin@localhost:~/Documents/School/Art$ cd ..
sysadmin@localhost:~/Documents/School$
```

Shortcut: The .. Characters



to move from the `School` directory to the `Downloads` directory

```
sysadmin@localhost:~/Documents/School$ cd ../../Downloads
sysadmin@localhost:~/Downloads$
```

Shortcut: The . Characters

Regardless of which directory the user is in, the single period `.` character always represents the current directory. For the `cd` this shortcut is not very useful.

Recursive Listing

ls –R

```
sysadmin@localhost:~$ ls -R /etc/ppp
/etc/ppp:
ip-down.d  ip-up.d

/etc/ppp/ip-down.d:
bind9

/etc/ppp/ip-up.d:
bind9
```

Sort Listing

ls –lS
ls –lt
ls –lh

ls –rlS
ls –rlt
ls –rlh